

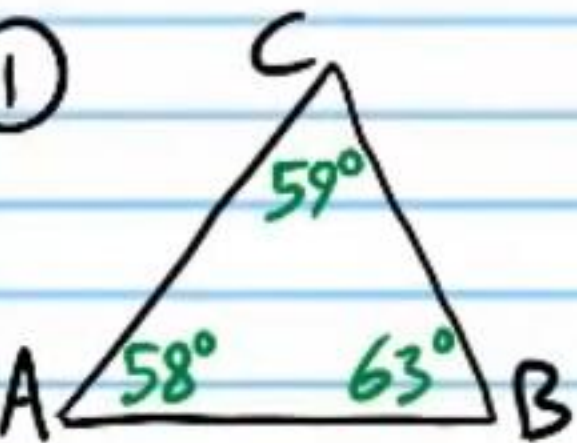
Ratios, Proportions, and Triangle Inequality Theorems

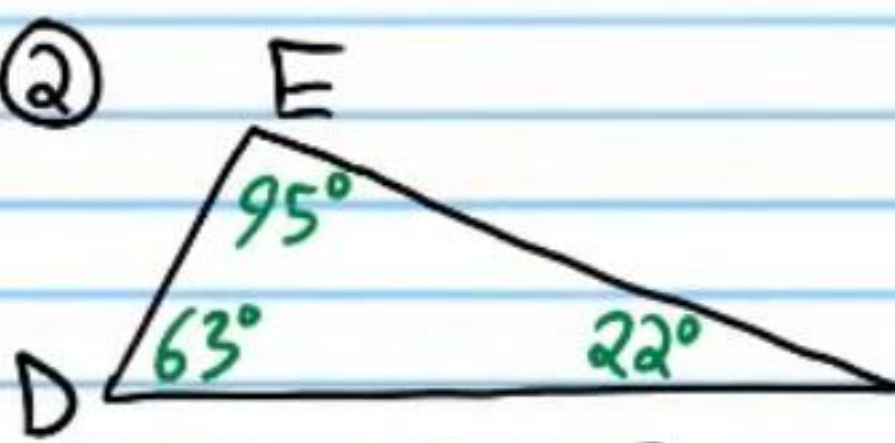
Triangle Inequality Theorems

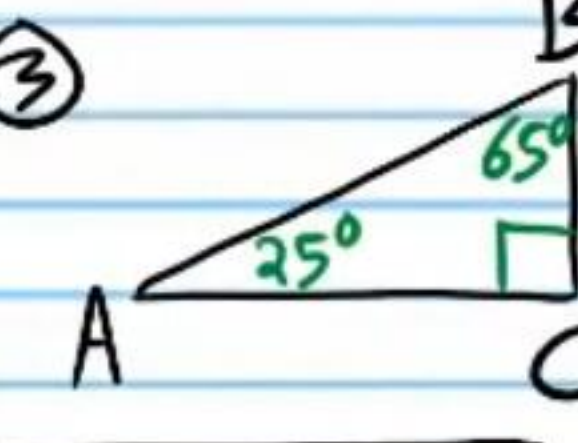
Triangle Side Inequality Theorem: If one angle in a triangle is bigger than another angle, then the side opposite the bigger angle is longer than the side opposite the smaller angle.

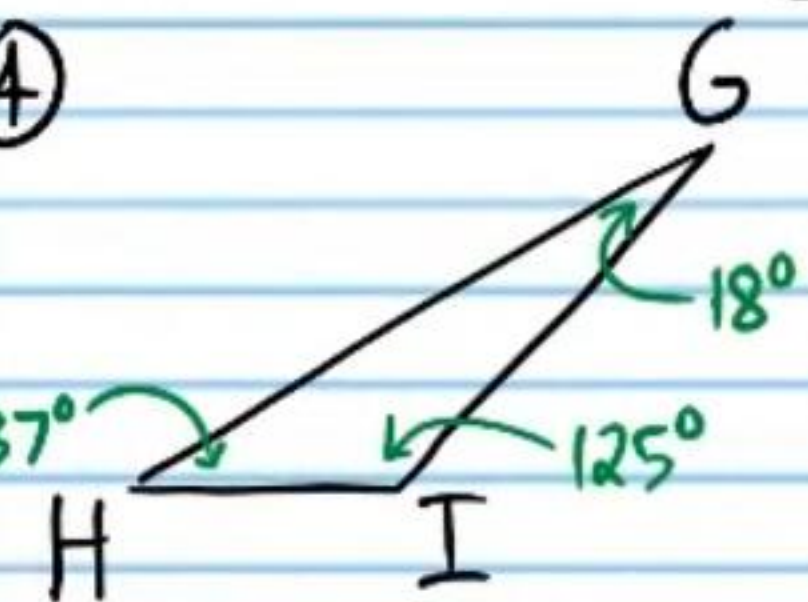
Triangle Angle Inequality Theorem: If one side of a triangle is longer than another side, then the angle opposite the longer side is bigger than the angle opposite the shorter side.

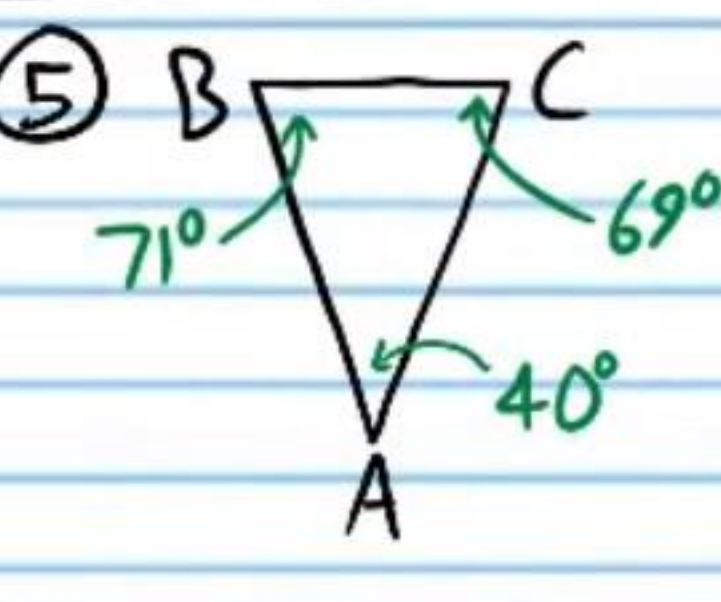
List the sides of the triangles from shortest to longest.

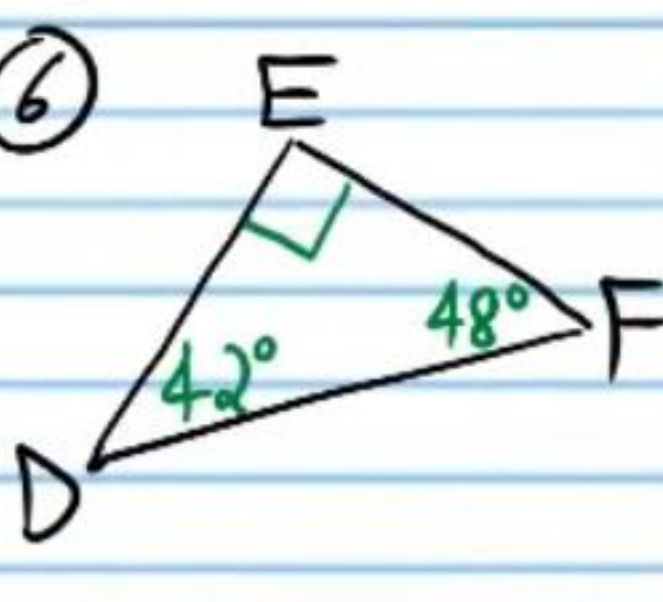
① 
 $\angle C, \angle A, \angle B$

② 
 $\angle E, \angle F, \angle D$

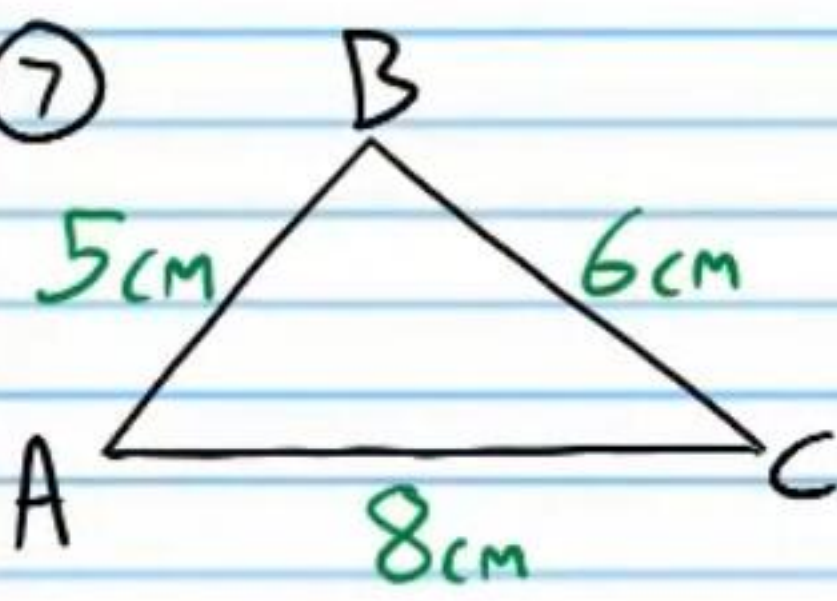
③ 
 $\angle C, \angle A, \angle B$

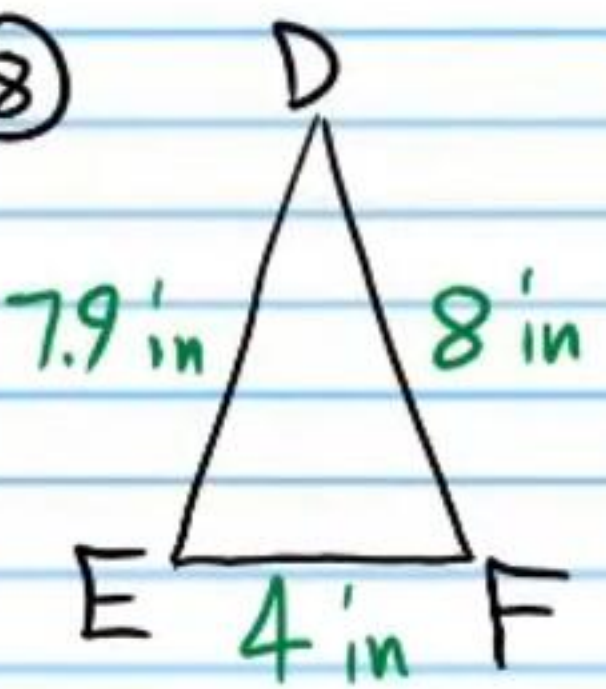
④ 
 $\angle I, \angle G, \angle H$

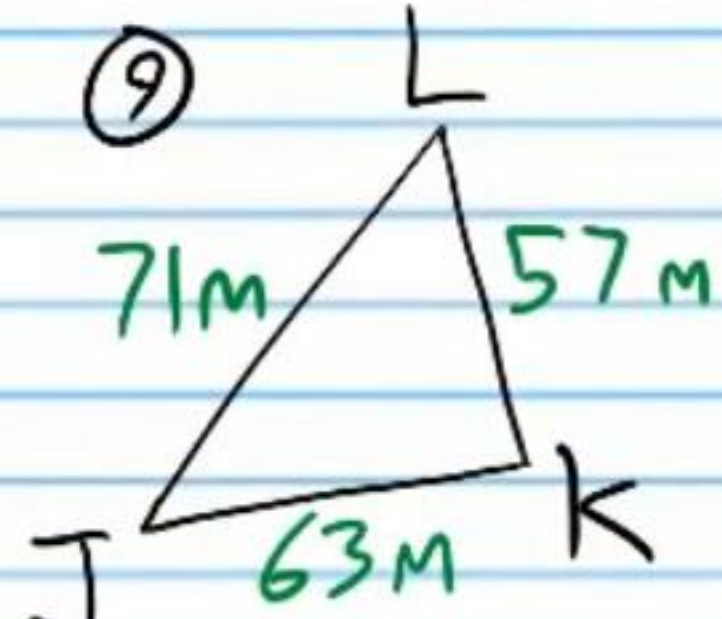
⑤ 
 $\angle C, \angle B, \angle A$

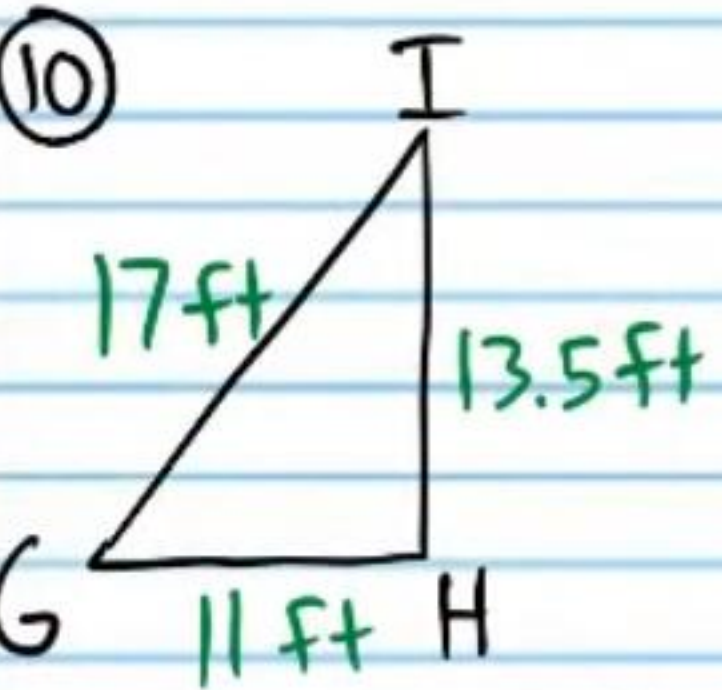
⑥ 
 $\angle F, \angle E, \angle D$

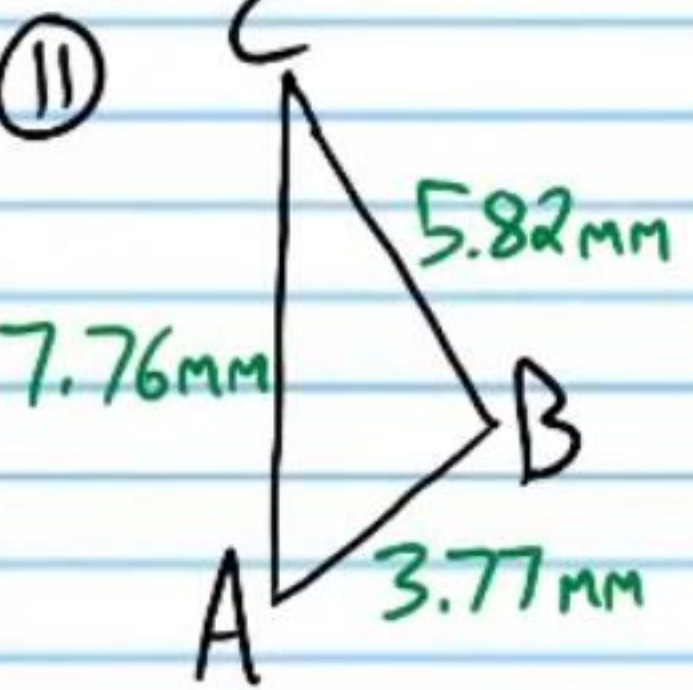
List the angles of the triangles from least to greatest.

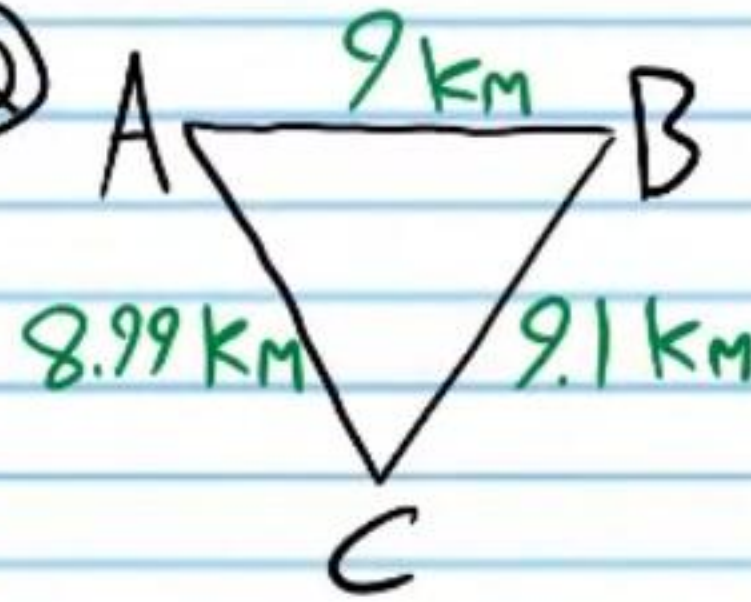
⑦ 
 $\angle C, \angle A, \angle B$

⑧ 
 $\angle D, \angle F, \angle E$

⑨ 
 $\angle J, \angle L, \angle K$

⑩ 
 $\angle I, \angle G, \angle H$

⑪ 
 $\angle C, \angle A, \angle B$

⑫ 
 $\angle B, \angle C, \angle A$

Review on Inequalities

Solve the following inequalities.

⑬ $3 + 2x > 9$
 $2x > 6$
 $x > 3$

⑭ $5 \leq -\frac{x}{4} + 1$
 $4 \leq -\frac{x}{4}$
 $-16 \geq x$

⑮ $-7x + 2 + x > 9$
 $-6x + 2 > 9$
 $-6x > 7$
 $x < -\frac{7}{6}$

$$(16) -2(x+1)-4 < 7+8(x-2)$$

$$-2x-2-4 < 7+8x-16$$

$$-2x-6 < -9+8x$$

$$-6 < -9+10x$$

$$3 < 10x$$

$$\boxed{\frac{3}{10} < x}$$

$$(17) -4x+1 \geq 9$$

$$-4x \geq 8$$

$$\boxed{x \leq -2}$$

$$(18) \frac{x}{7} - 2 > 4 \quad (19) 4 \geq 3x - 4x + 8 \quad (20) 3 + 2(3x - 4) < -7$$

$$\frac{x}{7} > 6$$

$$\boxed{x > 42}$$

$$4 \geq -x + 8$$

$$-4 \geq -x$$

$$\boxed{4 \leq x}$$

$$3 + 6x - 8 < -7$$

$$6x - 5 < -7$$

$$6x < -2$$

$$\boxed{x < -\frac{1}{3}}$$

Triangle Inequality Theorem: The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

Is it possible to make a triangle with each group of side lengths below? If not, write the inequality that shows that it's not possible.

$$(21) 10, 3, 8$$

$$\text{I) } 10 + 3 > 8$$

$$13 > 8$$

$$\text{II) } 10 + 8 > 3$$

$$18 > 3$$

$$\text{III) } 3 + 8 > 10$$

$$11 > 10$$

Yes

$$(22) 5, 2, 2$$

$$\text{I) } 5 + 2 > 2$$

$$7 > 2$$

$$\text{II) } 2 + 2 > 5$$

$$4 > 5$$

No

$$(23) 8, 7, 6$$

$$\text{I) } 8 + 7 > 6$$

$$15 > 6$$

$$\text{II) } 8 + 6 > 7$$

$$14 > 7$$

$$\text{III) } 7 + 6 > 8$$

$$13 > 8$$

Yes

$$(24) 9, 5, 4$$

$$\text{I) } 9 + 5 > 4$$

$$14 > 4$$

$$\text{II) } 9 + 4 > 5$$

$$13 > 5$$

$$\text{III) } 5 + 4 > 9$$

$$9 > 9$$

No

$$(25) 6, 2, 9$$

$$\text{I) } 6 + 2 > 9$$

$$8 > 9$$

No

$$(26) 4, 10, 6$$

$$\text{I) } 4 + 10 > 6$$

$$14 > 6$$

$$\text{II) } 4 + 6 > 10$$

$$10 > 10$$

No

$$(27) 13, 5, 12$$

$$\text{I) } 13 + 5 > 12$$

$$18 > 12$$

$$\text{II) } 13 + 12 > 5$$

$$25 > 5$$

$$\text{III) } 5 + 12 > 13$$

$$17 > 13$$

Yes

$$(28) 4, 1, 4$$

$$\text{I) } 4 + 1 > 4$$

$$5 > 4$$

$$\text{II) } 4 + 4 > 1$$

$$8 > 1$$

Yes

If each pair of numbers below represents side lengths of a triangle, write an inequality that shows the possible lengths of the third side X .

②⑨ 12, 7

Side lengths

12, 7, X

I) $12 + 7 > X$

$19 > X$

II) $12 + X > 7$

$X > -5$

III) $7 + X > 12$

$X > 5$

Final Inequality

$5 < X < 19$

③⑩ 10, 13

Side Lengths

10, 13, X

I) $10 + 13 > X$

$23 > X$

II) $10 + X > 13$

$X > 3$

Final Inequality

$3 < X < 23$

③⑪ 6, 2

Side Lengths

6, 2, X

I) $6 + 2 > X$

$8 > X$

II) $6 + X > 2$

$X > -4$

III) $2 + X > 6$

$X > 4$

Final Inequality

$4 < X < 8$

③⑫ 11, 5

Side Lengths

11, 5, X

I) $11 + 5 > X$

$16 > X$

II) $5 + X > 11$

$X > 6$

Final Inequality

$6 < X < 16$

③⑬ 2, 4

Side Lengths

2, 4, X

I) $2 + 4 > X$

$6 > X$

II) $2 + X > 4$

$X > 2$

Final Inequality

$2 < X < 6$

③⑭ 27, 80

Side Lengths

27, 80, X

I) $27 + 80 > X$

$107 > X$

II) $27 + X > 80$

$X > 53$

Final Inequality

$53 < X < 107$

Ratios

Example: Don jumped 4 feet and Tim jumped 3 feet.

Three Ways to Write a Ratio

i) $\frac{4}{3}$

ii) 4:3

iii) 4 to 3

35) I have one dollar and you have six dollars. Write ratios of these two numbers in six different ways.

$$\frac{1}{6}, 1:6, 1 \text{ to } 6, \frac{6}{1}, 6:1, 6 \text{ to } 1$$

36) Billy ate 20 sandwiches last month and Tom ate 11 sandwiches last month. Write ratios of these two numbers in six different ways.

$$20/11, 20:11, 20 \text{ to } 11, \frac{11}{20}, 11:20, 11 \text{ to } 20$$

37) The Kawasaki H2 supercharged motorcycle has 300 horsepower, while the ZX-6R motorcycle has 107 horsepower. Write ratios of these two numbers in six different ways.

$$\frac{300}{107}, 300:107, 300 \text{ to } 107, \frac{107}{300}, 107:300, 107 \text{ to } 300$$

Reducing Ratios to Lowest Terms

38) There were 8 gumballs in one bowl and 6 gumballs in another bowl. Write these two numbers as a ratio in lowest terms using a colon. Then, write the simplified ratio in two other ways.

$$8:6 \Rightarrow 4:3, \frac{4}{3}, 4 \text{ to } 3$$

39) A store sells a microphone for \$63, while another store sells the same model microphone for \$49. Write a ratio of these two numbers in lowest terms using a colon. Then, write this simplified ratio in two other ways.

$$63:49 \Rightarrow 9:7, \frac{9}{7}, 9 \text{ to } 7$$

Write the following ratios in lowest terms.

40) 45:25

$$9:5$$

41) 10 to 24

$$5 \text{ to } 12$$

42) $\frac{36}{6}$

$$6/1$$

43) $\frac{28}{21}$

$$\frac{4}{3}$$

44) 10 to 20

$$1 \text{ to } 2$$

45) 26:26

$$1:1$$

Proportions

Determine if the proportions are true.

$$(46) \frac{3}{5} \stackrel{?}{=} \frac{12}{20}$$

$$\frac{12}{20} = \frac{12}{20}$$

Yes

$$(47) \frac{10}{22} \stackrel{?}{=} \frac{1}{2}$$

$$\frac{10}{22} \neq \frac{11}{22}$$

No

$$(48) \frac{4}{6} \stackrel{?}{=} \frac{2}{3}$$

$$\frac{2}{3} = \frac{2}{3}$$

Yes

$$(49) \frac{3}{5} \stackrel{?}{=} \frac{8}{18}$$

$$\frac{54}{90} \neq \frac{40}{90}$$

No

$$(50) \frac{10}{12} \stackrel{?}{=} \frac{40}{45}$$

$$\frac{5}{6} \neq \frac{8}{9}$$

$$\frac{15}{18} \neq \frac{16}{18}$$

No

$$(51) \frac{20}{35} \stackrel{?}{=} \frac{8}{14}$$

$$\frac{4}{7} = \frac{4}{7}$$

Yes

$$(52) \frac{12}{16} \stackrel{?}{=} \frac{9}{12}$$

$$\frac{3}{4} = \frac{3}{4}$$

Yes

$$(53) \frac{72}{81} \stackrel{?}{=} \frac{21}{27}$$

$$\frac{8}{9} \neq \frac{7}{9}$$

No

$$(54) \frac{30}{6} \stackrel{?}{=} \frac{14}{3}$$

$$\frac{5}{1} \neq \frac{14}{3}$$

No

$$(55) \frac{9}{3} \stackrel{?}{=} \frac{20}{5}$$

$$\frac{3}{1} \neq \frac{4}{1}$$

No

Solve the Proportions

$$(56) \frac{8}{n} = \frac{24}{27}$$

$$\frac{8}{n} = \frac{8}{9}$$

$$n = 9$$

$$(57) \frac{3}{15} = \frac{a}{5}$$

$$\frac{1}{5} = \frac{a}{5}$$

$$a = 1$$

$$(58) \frac{42}{63} = \frac{6}{b}$$

$$\frac{6}{9} = \frac{6}{b}$$

$$9 = b$$

$$(59) \frac{n}{3} = \frac{12}{18}$$

$$\frac{n}{3} = \frac{2}{3}$$

$$n = 2$$

(60) If a bird flies 22 miles in 55 minutes, how long would it take this bird to fly 2 miles? Write a proportion statement and use it to solve this problem.

$$\frac{22}{55} = \frac{2}{h}$$

$$\frac{2}{5} = \frac{2}{h}$$

$$h = 5 \text{ hours}$$

(61) If 40 calories of energy are required to run 90 feet, how many calories would be required to fly 9 feet? Write a proportion statement and use it to solve this problem.

$$\frac{40}{90} = \frac{c}{9}$$

$$\frac{4}{9} = \frac{c}{9}$$

$$c = 4 \text{ cal}$$

$$\textcircled{62} \frac{9}{4} = \frac{18}{n}$$

$$\frac{18}{8} = \frac{18}{n}$$

$$\boxed{n=8}$$

$\textcircled{63}$ If there are 5 ounces of silver for every 2 ounces of gold, how many ounces of gold would be present with 30 ounces of silver? Write a proportion statement and use it to solve this problem.

$$\frac{5}{2} = \frac{30}{g}$$

$$\frac{30}{12} = \frac{30}{g}$$

$$\boxed{g=12 \text{ oz}}$$

$$\textcircled{64} \frac{b}{14} = \frac{8}{7}$$

$$\frac{b}{14} = \frac{16}{14}$$

$$\boxed{b=16}$$

$\textcircled{65}$ If there are 2 gallons of red paint for every 7 gallons of white paint, how many gallons of red paint would be present with 21 gallons of white paint? Write a proportion statement and use it to solve this problem.

$$\frac{2}{7} = \frac{R}{21}$$

$$\frac{6}{21} = \frac{R}{21}$$

$$\boxed{R=6 \text{ gal}}$$

$$\textcircled{66} \frac{100}{n} = \frac{10}{9}$$

$$\frac{100}{n} = \frac{100}{90}$$

$$\boxed{n=90}$$

$$\textcircled{67} \frac{n}{5} = \frac{7}{8}$$

$$8n = 35$$

$$n = \frac{35}{8}$$

$$\boxed{n = 4 \frac{3}{8}}$$

$$\textcircled{68} \frac{4}{5} = \frac{6}{a}$$

$$4a = 30$$

$$a = \frac{30}{4}$$

$$\boxed{a = 7 \frac{1}{2}}$$

$$\textcircled{69} \frac{7}{2} = \frac{b}{3}$$

$$2b = 21$$

$$b = \frac{21}{2}$$

$$\boxed{b = 10 \frac{1}{2} \text{ or } 10.5}$$

$$\textcircled{70} \frac{2}{c} = \frac{5}{9}$$

$$5c = 18$$

$$c = \frac{18}{5}$$

$$\boxed{c = 3 \frac{3}{5}}$$